Systems Analyst Assignments: E-commerce System

Systems Analysis and Design | Given System: E-commerce System

Assignment 1: Skills of a Systems Analyst

An E-commerce system lets customers browse products, add items to a cart, pay online, and track deliveries, while sellers and admins manage inventory, orders, and payments. A systems analyst needs four main skills to build it successfully.

1. Interpersonal Skills

These help the analyst communicate and work with different people. In an e-commerce project, the analyst must talk to customers, sellers, warehouse staff, delivery partners, marketing teams, and business owners. Each group has different needs and a different level of technical knowledge. Good listening, clear speaking, and patience help the analyst understand their problems and explain the system in simple words. Teamwork and negotiation also help resolve conflicts, such as when the marketing team wants many discount features but the finance team wants to control costs.

2. Technical Skills

Technical skills let the analyst understand how the system will actually be built. For an e-commerce system, the analyst should know database design (products, orders, customers), web technologies, networking, and payment gateway integration. Since customers enter card and personal data, the analyst must also understand security topics such as encryption, HTTPS, and secure login. These skills help the analyst talk with developers and choose suitable tools such as MySQL, PHP, React, or Node.js.

3. Analytical Skills

Analytical skills mean the ability to study a problem, break it into parts, and find the best solution. The analyst studies the current selling process, such as phone orders or social media sales, and finds its problems: manual order tracking, stock mismatches, payment confusion, and delivery delays. Tools like DFDs, flowcharts, and use case diagrams help model the process. The analyst then suggests how the new system can fix these problems.

4. Management Skills

Management skills help the analyst plan and control the project. In an e-commerce project, the analyst must manage time, budget, and resources, and divide tasks among team members. Risk management is also important, for example handling website crashes during sales campaigns or payment failures. Good management ensures the project finishes on time and users accept the new platform.

Assignment 2: Methodologies for Requirement Determination

Requirement determination means finding out what the users and the organization need from the new system. If requirements are collected wrongly, the final system will fail. The following methodologies are used, following the six steps shown in the slide.

Step 1: Stakeholder Identification

First, we identify everyone affected by the e-commerce system: customers, sellers, admins, warehouse staff, delivery partners, customer support, payment providers, and business owners. Each stakeholder has different expectations, so none should be missed.

Step 2: Requirement Elicitation Techniques

This is the step where requirements are actually collected. Common techniques are:

- **Interviews:** One-to-one talks with sellers and business owners give detailed and accurate information, but they take a lot of time.
- **Questionnaires:** Useful for collecting opinions from many customers quickly, such as preferred payment methods, but answers may be shallow.
- **Observation:** The analyst watches how orders are packed and shipped in the warehouse. It shows hidden problems, but people may behave differently when watched.
- **Document Analysis:** Studying old order records, invoices, and competitor websites helps in understanding data and common features.
- **Prototyping:** A sample checkout page is shown to users so they can give early feedback.
- **JAD (Joint Application Development):** Users and developers meet in workshops to decide requirements together, which saves time.

Step 3: Requirement Documentation

All collected requirements are written in an SRS (Software Requirements Specification), divided into functional requirements (e.g., search products, place orders, make payments) and non-functional requirements (e.g., security, fast page loading, handling many users at once).

Step 4: Requirements Analysis and Prioritization

Requirements are checked for conflicts and duplication. They are then ranked, for example using the MoSCoW method. Product search, cart, and secure payment are "must haves," while a wishlist or product recommendation may be a "could have."

Step 5: Verification and Validation

The team checks that requirements are complete and correct (verification) and that they really match user needs (validation). Reviews, walkthroughs, and prototype demos are used here.

Step 6: Requirements Management and Change Control

Requirements often change, such as a new payment method like bKash or a new delivery policy. A change control process records, reviews, and approves each change so the project does not go out of control.

Conclusion

For an e-commerce system, no single technique is enough. A combination of interviews, questionnaires, observation, and prototyping gives the best result. Following all six steps carefully helps the analyst collect clear, correct requirements and build a system that meets both customer and business needs.